Mines Paris PSL - Etilab

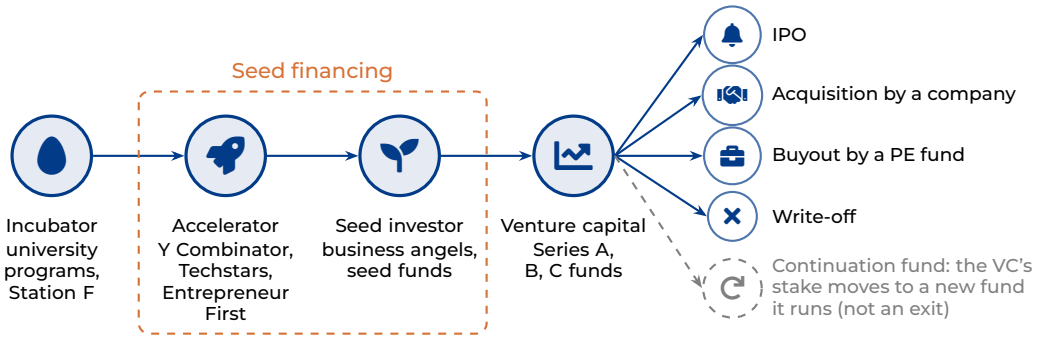
Protection and Dilution: SAFEs and MFN Clauses in Seed Financing

Stephen Dossou
Under the supervision of
Pierre Fleckinger and Philippe Mustar

Imperial College Business School, 12 November 2026

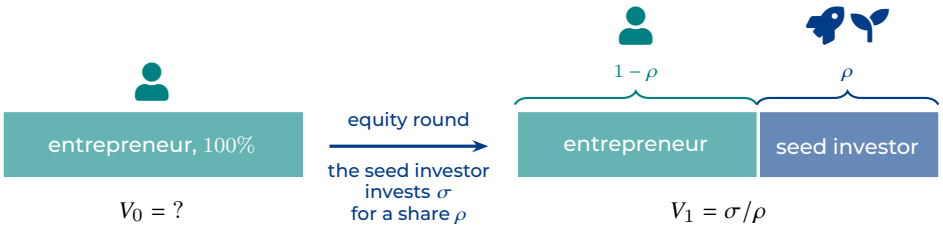
The life of a venture-backed startup

A startup raises money from a sequence of investors before it exits.



Exit routes follow Cumming and MacIntosh (2003): IPO, acquisition, secondary sale, buyback, write-off.

A first equity round combines the first issuance of N shares at price p.

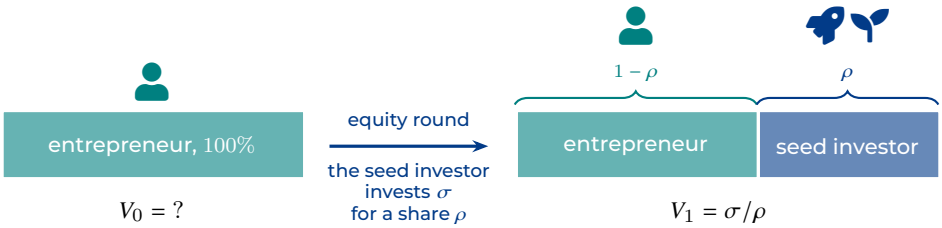


Definition

An equity round is characterized by (σ, ρ, N, p) such that $Np = \sigma/\rho = V_1$.

A seed investor invests \$120K for 7% equity, so $V_1 = \$120\text{K}/0.07 \approx \$1.71\text{M}$.

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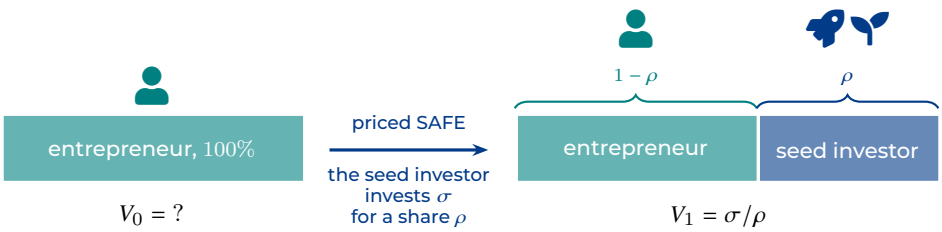
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But equity rounds are administratively heavy (new share class, legal fees).

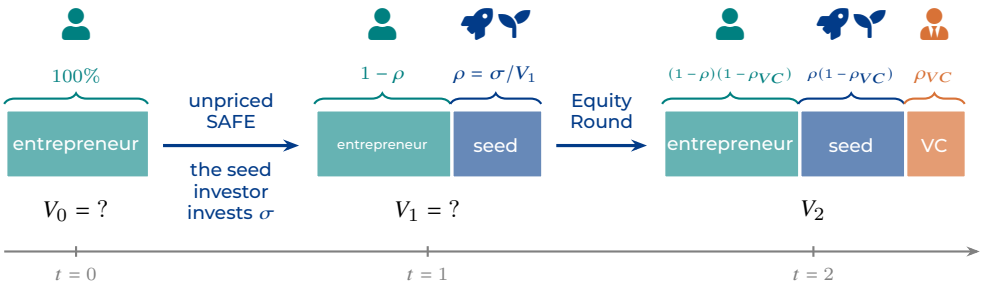
Contracts in seed financing
Priced SAFE (Simple Agreement for Future Equity)

A first priced SAFE defines the project's first valuation without issuance of shares.

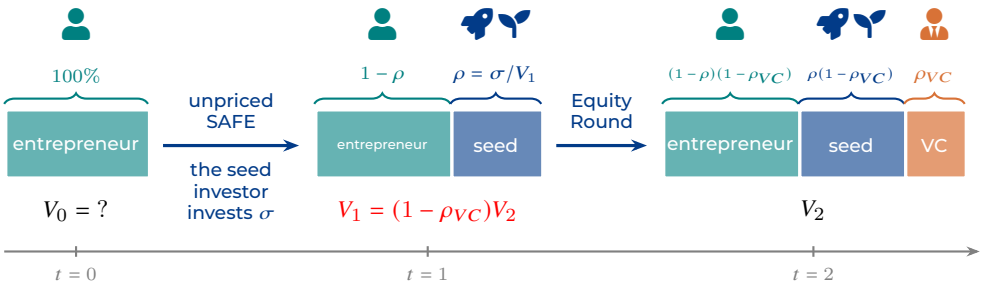


A seed investor invests \$125K for 7% on a SAFE, so $V_1 = \$125\text{K}/0.07 \approx \$1.79\text{M}$.

A first unpriced SAFE with no cap and no discount leaves the valuation to the next round.

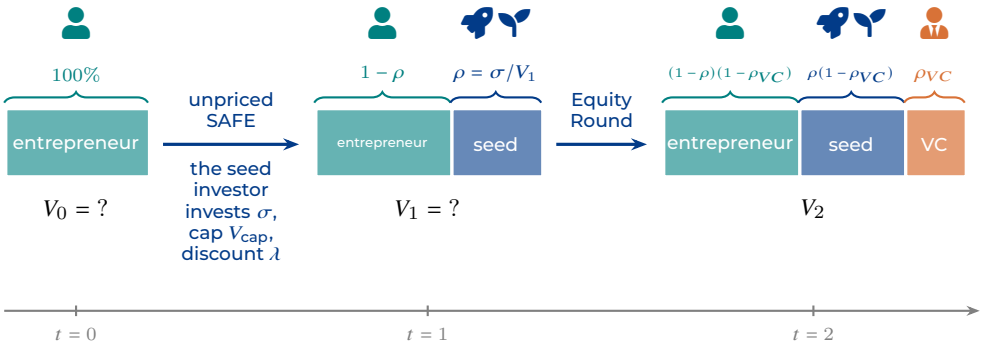


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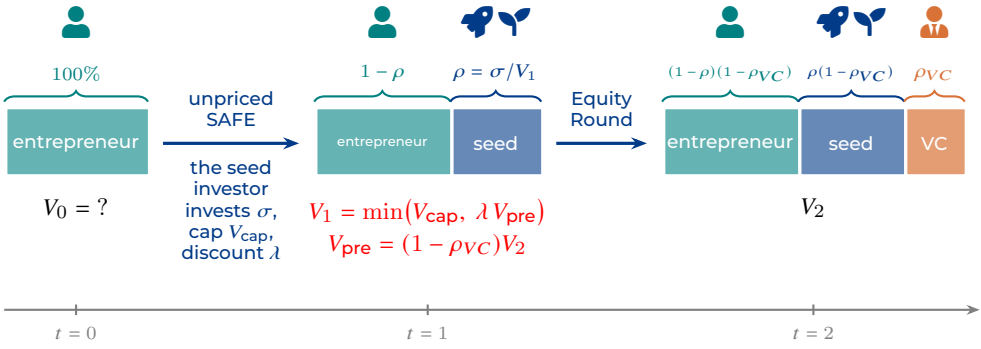
Contracts in seed financing
Unpriced SAFE with a cap V_{cap} and a discount λ

A first unpriced SAFE defines a valuation cap and a discount without issuance of shares.

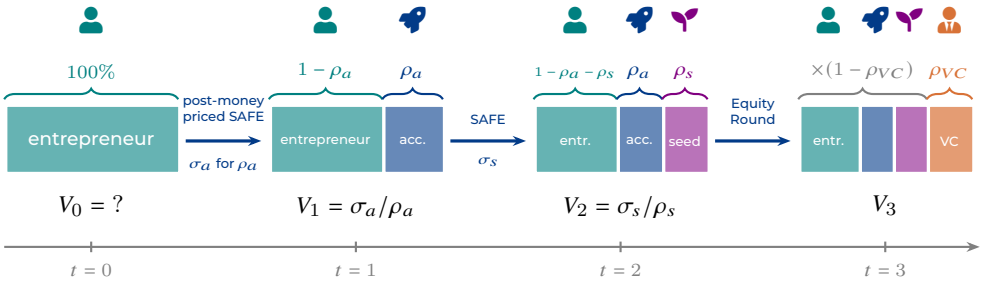


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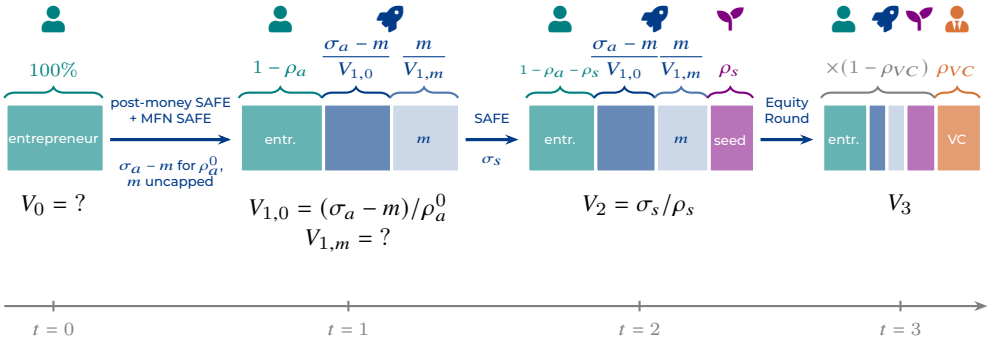


The accelerator's post-money priced SAFE fixes its share ρ_a until the equity round.



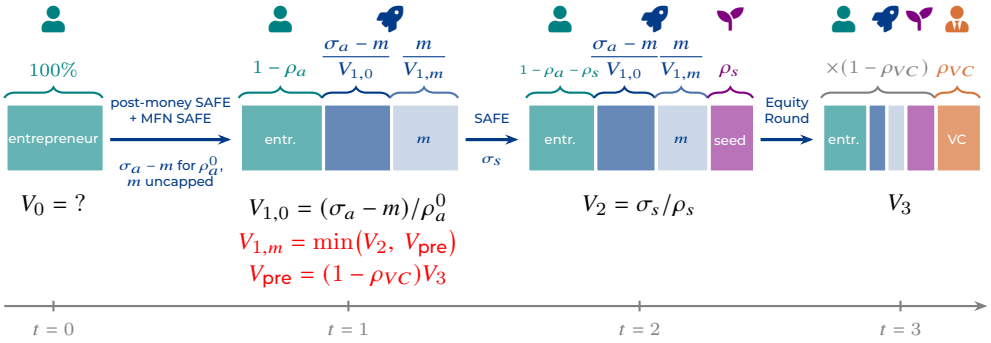
"Post-money" ensures that other SAFEs cannot dilute the accelerator's SAFE.

The accelerator can add to its post-money SAFE an MFN SAFE, which converts on the best upcoming terms.



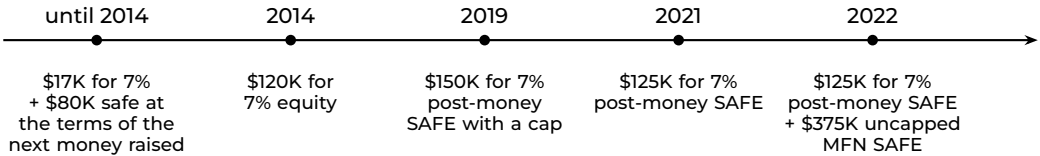
The accelerator's SAFEs convert at two valuations. The MFN SAFE buys $m/V_{1,m}$: it takes more from a seed round with a low valuation.

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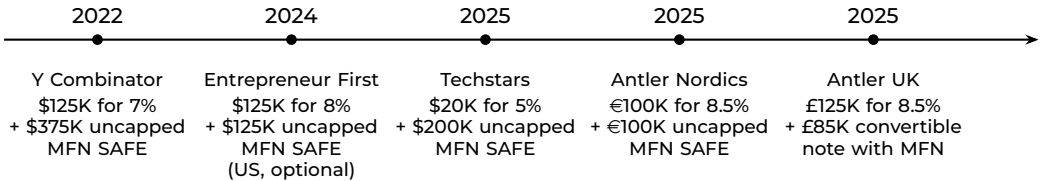
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Chronology of Y Combinator deal terms



Accelerators' MFN SAFEs

Y Combinator launched a trend among accelerators.



A convertible note works like a SAFE, but it is debt: it carries an interest rate and a maturity, and ranks senior in liquidation.

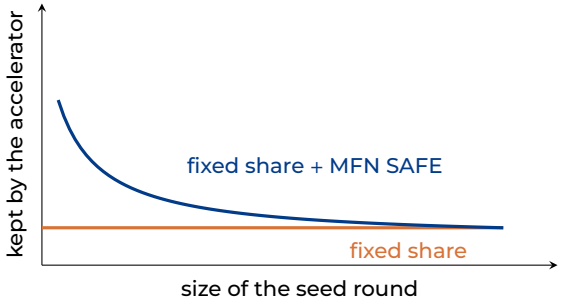
The question

Research question

Is it worth adding an MFN SAFE to a post-money priced SAFE?

- For the accelerator: does the clause raise its profit, and which protection should it choose?
- For the projects: does it raise the probability of a seed round? Of an equity round?
- Across projects: how do these anti-dilution clauses separate software and deep-tech projects?

The idea: an MFN SAFE takes more from small seed rounds



- The MFN SAFE converts at the seed investor's price. A small round sets a low price, so the same amount buys a larger share.
- A fixed share takes the same from every round: to protect more, it must give up the largest rounds.

What is known, and what this paper adds

Literature	Key papers	What we know	What this paper adds
Experimen-tation	Kerr, Nanda, Rhodes-Kropf (2014) Ewens, Nanda, Rhodes-Kropf (2018) Bergemann, Hege (1998, 2005)	Value is learned by investing, so early money funds an experiment.	The terms of the second investor, once the first has invested.

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To our knowledge, the first model of an accelerator's protection against seed dilution.

Findings

Any fixed share with a contribution is beaten by a small MFN component.

For the accelerator

- It takes more from small rounds.
- The seed investor answers with an unpriced SAFE.
- And overinvests on small rounds.

For the projects

- A protection shortens the seed zone.
- The contribution moves it either way.
- Overinvestment raises full funding.

Software and deep tech

- A seed round covers at most $W(I) < I$.
- Software: the clause is best, and almost all the money goes through it.
- Beyond: a share for almost no money, a quarter of the surplus lost.

The agents

The project

- A liquidity shock $\ell \sim g$.
- A scaling cost I.
- A value $\pi \sim f$.

(A1) $E(\pi) < I$.
(A2) f, g positive, log-concave.

The accelerator

- Owns the whole project.
- Spends σ_a before ℓ and π.
- Can add anti-dilution clauses.

The seed investor

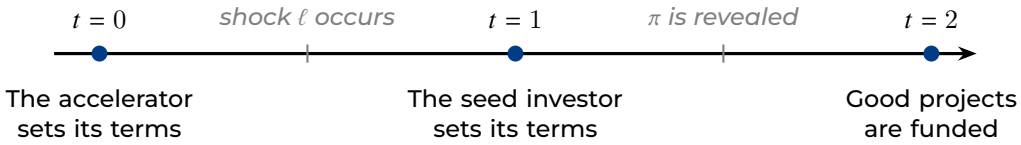
- Invests σ_s after ℓ, before π.
- Makes TIOLI offers.
- Runs an equity round, or a priced or unpriced SAFE.

Experimentation

Covering the liquidity shock ℓ finances an experiment, which reveals the project's value π. The accelerator and the seed investor are the investors able to finance it.

▸ An example

Timing



- The accelerator's contribution is spent before the shock: if it rejects the seed round at $t = 1$, the project is abandoned and it gets nothing.
- If the project is not funded at $t = 2$, it returns nothing, and all the money invested is lost.

We solve the game by backward induction.

Throughout

These hold in every result that follows.

- (A1) $E(\pi) < I$ and (A2) f, g positive and log-concave.
- The seed investor enters only when the shock exceeds the contribution, $\ell > \sigma_a$, and the remaining need stays in $[0, 1]$, $x \in [0, 1]$.
- Every SAFE converts at full funding, at the pre-money value $\pi - x$.
- The accelerator's protection comes first. As in YC's post-money SAFE, it is not diluted by the SAFEs signed after it, and the later SAFEs share what it leaves.

Benchmark: no clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds no protection. Seed holds a priced SAFE.

Suppose $\ell > \sigma_a$ and $x \in [0, 1]$. The seed investor solves

$$\begin{aligned} \max_{\sigma_s, \rho_s} \quad & \Pi_s = \rho_s\, W(x) - \sigma_s \\ \text{under} \quad & \sigma_s \geq \ell - \sigma_a \qquad (L) \end{aligned}$$

with $\rho_s \in [0, 1]$.

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Accelerator holds no protection. Seed holds a priced SAFE.

- The seed investor exactly covers the remaining liquidity cost, $\sigma_s^* = \ell - \sigma_a$, and $x^* = I$.
- It takes all of the equity, $\rho_s^* = 1$.
- It earns $\Pi_s^* = W(I) - (\ell - \sigma_a)$, and finances $\ell \leq \sigma_a + W(I)$.
- The accelerator keeps nothing.

Benchmark: no clause, priced SAFE

The project is fully funded when $\pi > x$, with $x = I + \ell - \sigma_a - \sigma_s$ the remaining need after the seed stage, and $W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi$ the surplus.
Accelerator holds no protection. Seed holds a priced SAFE.

The accelerator's contribution is spent before the shock. It solves

$$\max_{\sigma_a \geq 0} \quad \Pi_a = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a.$$

Benchmark: no clause, priced SAFE

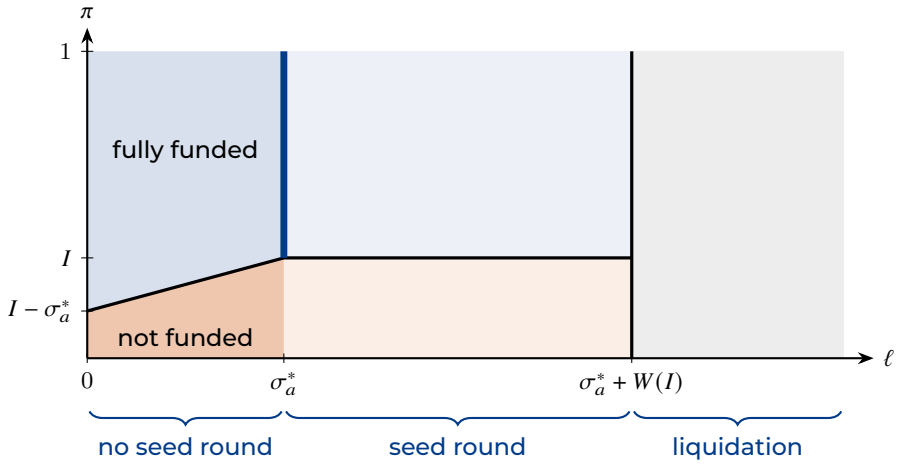
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Accelerator holds no protection. Seed holds a priced SAFE.

Every optimal contribution σ_a^* is zero or solves

$$W(I)\, g(\sigma_a^*) = \big(1 - G(\sigma_a^*)\big) + \int_0^{\sigma_a^*} F(I + \ell - \sigma_a^*)\, g(\ell)\, d\ell.$$

Benchmark: no clause, priced SAFE



Setup | Uncapped SAFE | With discount | Capped SAFE

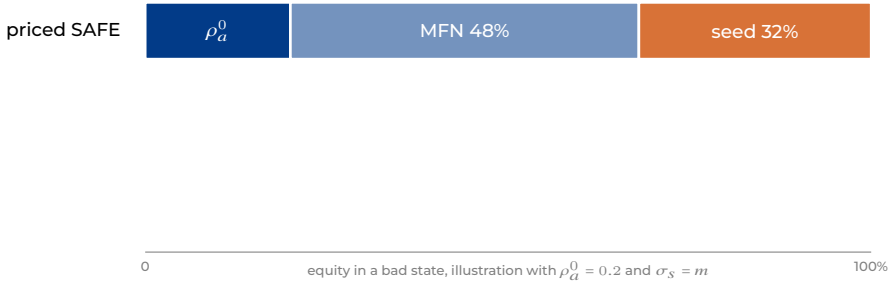
The seed investor's answer to each protection

Accelerator	Last shock the seed investor finances	Accelerator keeps on each round
No clause	$\sigma_a + W(I)$	nothing
Fixed share ρ_a	$\sigma_a + (1-\rho_a)W(I)$	$\rho_a W(I)$
Own SAFE (V_a, λ_a)	$\sigma_a + W\big(I + \frac{\sigma_a}{\lambda_a}\big) - \frac{\sigma_a}{V_a} W\big(I + \frac{V_a}{\lambda_a}\big)$	$W(I) - W\big(I + \frac{\sigma_a}{\lambda_a}\big) + \frac{\sigma_a}{V_a} W\big(I + \frac{V_a}{\lambda_a}\big)$
MFN clause (ρ_a^0, m)	$\sigma_a + (1-\rho_a^0)W(I) - m$	at least $\rho_a^0 W(I)$, and $\rho_a^0 W(I) + m$ on the last round

Without the MFN clause, the seed investor invests its need, and the accelerator keeps the same value on every round.

▸ Fixed share, priced ▸ Fixed share, capped ▸ Own SAFE ▸ MFN clause

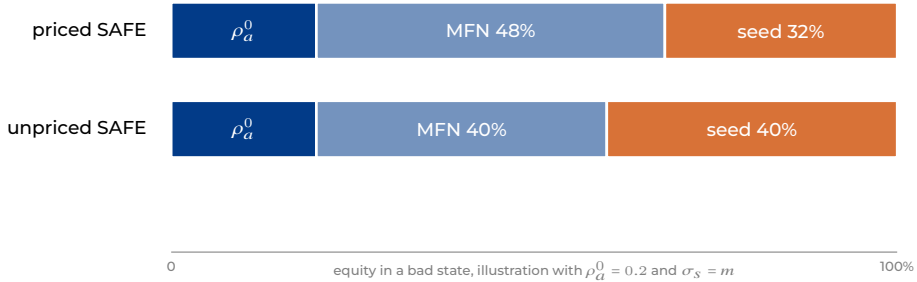
The seed investor prefers an unpriced SAFE



In a bad state, the equity round prices the project below the seed investor's valuation. The MFN money converts at this lower price: against a priced SAFE, it takes part of the seed investor's share.

▸ Proof

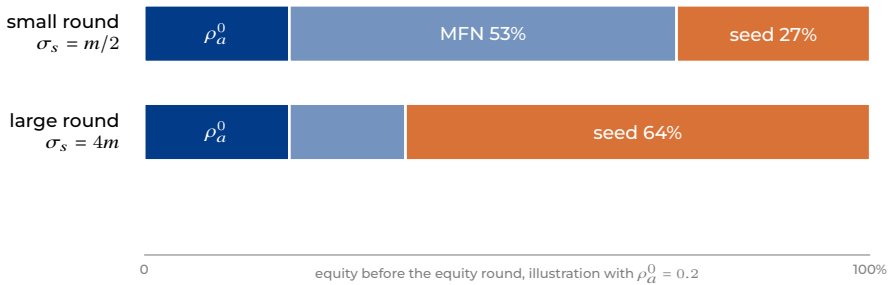
The seed investor prefers an unpriced SAFE



An unpriced SAFE converts at the same price as the MFN money, in every state: the seed investor keeps $(1 - \rho_a^0)\sigma_s/(\sigma_s + m)$, the most the clause allows. From now on, it uses an unpriced SAFE.

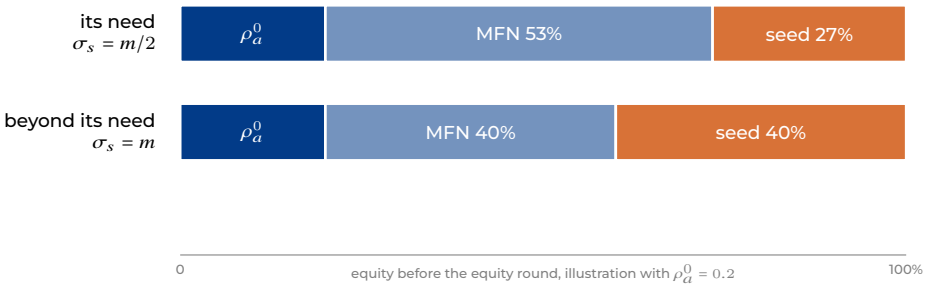
▸ Proof

On small rounds, the seed investor overinvests



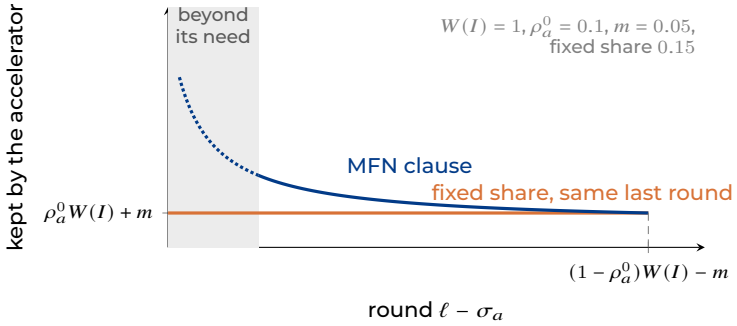
1. On a small round, the seed investor takes little.

On small rounds, the seed investor overinvests



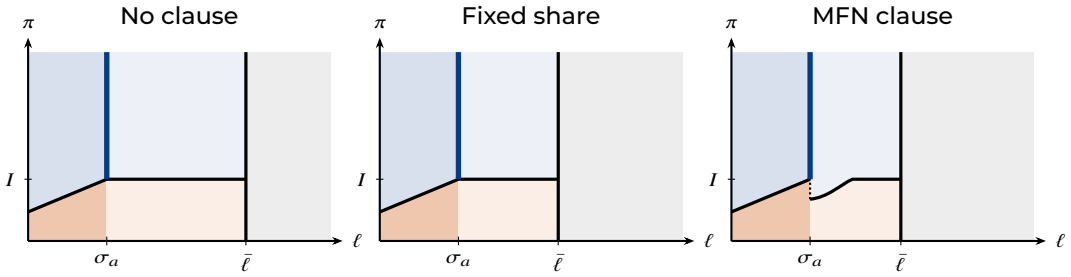
1. On a small round, the seed investor takes little.
2. It answers by investing beyond its need.

The MFN clause keeps more on every round but the last



With the same last round, the MFN clause keeps $\rho_a^0 W(I) + (1 - \rho_a^0)\, m\, W(I)/(\ell - \sigma_a + m)$. On the smallest rounds, the seed investor invests beyond its need to dilute the MFN amount.

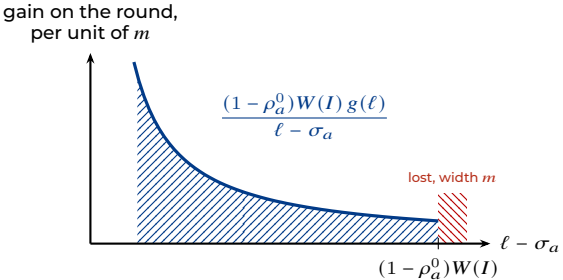
The MFN clause bends the seed round



Above the black line, the project is fully funded. Under the MFN clause, the seed investor's extra money lowers the threshold below I on the smallest rounds.

A small MFN component beats any fixed share

Take any fixed share with $\sigma_a > 0$ and $0 < \rho_a < 1$, and add a small MFN amount m. The profit strictly rises.



Per unit of m, the gain explodes on the smallest rounds. The loss, the last rounds of width m, stays bounded. So a small enough m raises the profit.

▸ Proof

Money beyond the need is lost

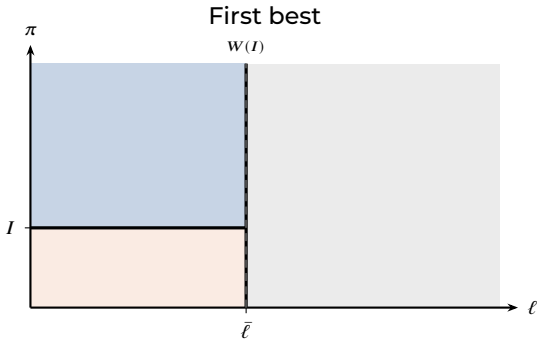
A covered project with threshold $x \leq I$ has surplus

$$W(x) - (I + \ell - x) \;=\; W(I) - \ell \;-\; \underbrace{(I - x)\, F(x)}_{\text{lost when } \pi < x} \;-\; \underbrace{\int_x^I (I - \pi)\, f(\pi)\, d\pi}_{\text{funds projects with } \pi < I} \;.$$

An abandoned project has surplus $-\sigma_a$: the contribution is spent before the shock.

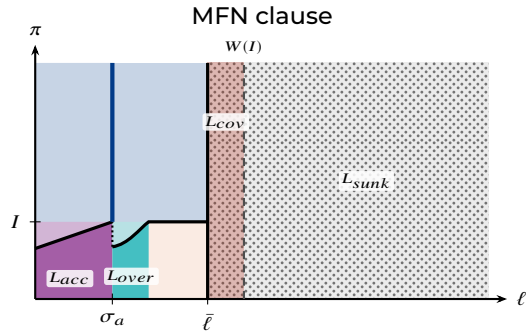
Every dollar brought beyond the need, $I - x$, is either lost or funds a project worth less than its scaling cost.

The first best brings no money before the shock



It covers exactly the shocks $\ell \leq W(I)$, at the threshold I: $TS^* = \int_0^{W(I)} \big(W(I) - \ell\big) g(\ell)\, d\ell$.
Without protection and without contribution, the seed investor reaches it.

Four losses from the first best

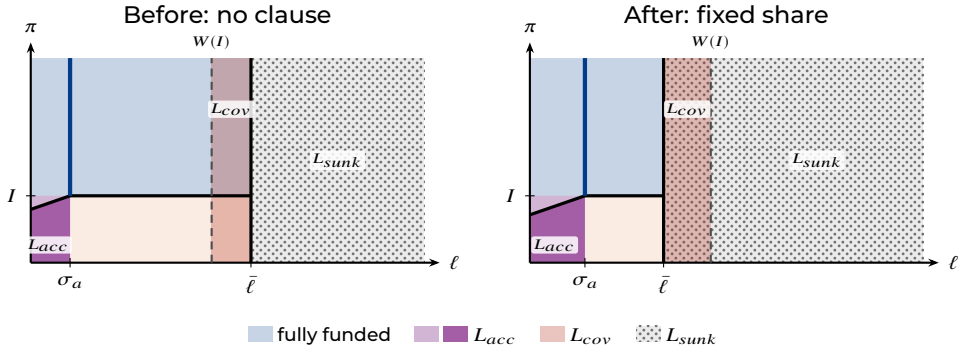


fully funded L_{acc} L_{over} L_{cov} L_{sunk}

$$TS = TS^* - L_{acc} - L_{over} - L_{cov} - L_{sunk}$$

▸ Derivation

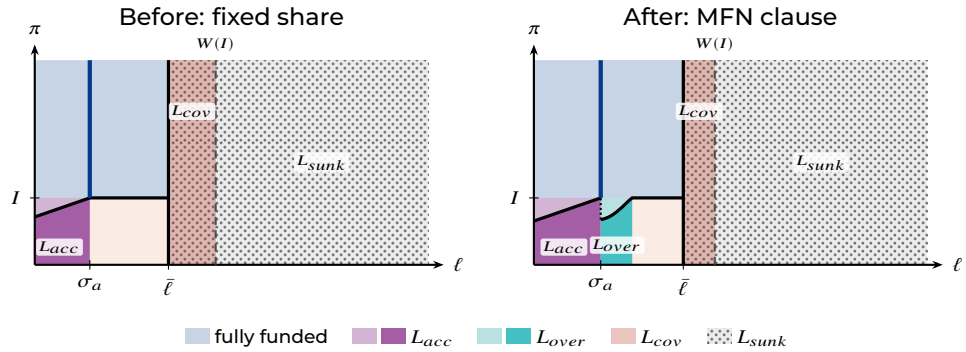
Total surplus, before and after a fixed share



The fixed share abandons the last shocks. Covering them was worth $\sigma_a + W(I) - \ell \geq 0$ more than abandoning, since the contribution is spent. Without protection, more contribution lowers the total surplus.

▸ Derivation

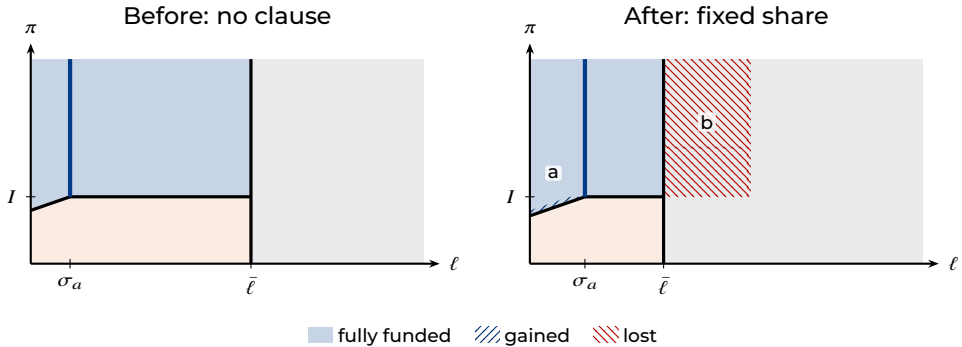
Total surplus, before and after the MFN clause



The MFN clause adds L_{over} on the smallest rounds. The last project and the contribution move the other losses either way.

▸ Derivation

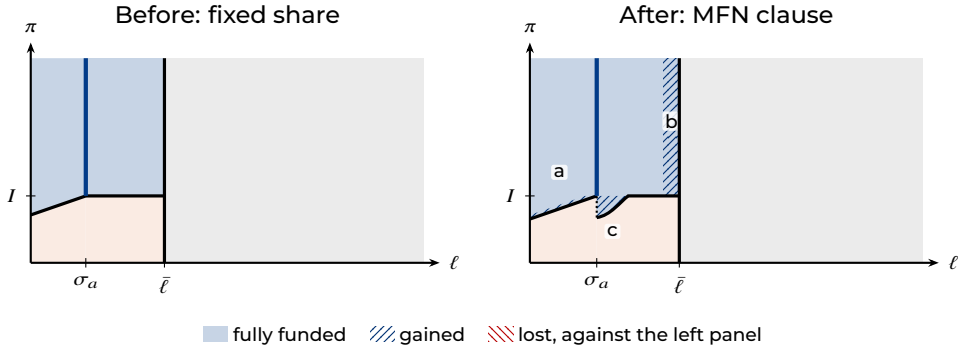
Full funding, before and after a fixed share



(a) The contribution moves either way, drawn higher here: more money lowers the threshold on the shocks the accelerator covers alone. (b) At a given contribution, the seed investor always stops earlier.

▸ Full funding ▸ Derivation

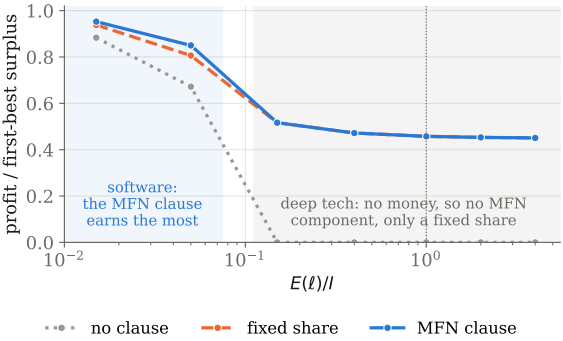
Full funding, before and after the MFN clause



(a) The contribution moves either way, drawn higher here. (b) The last project moves either way, drawn as a gain. (c) The money beyond the need always raises full funding.

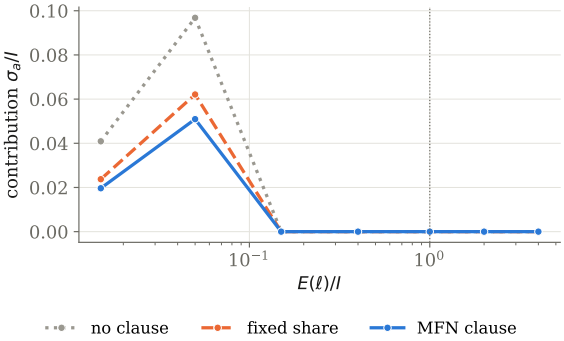
▸ Derivation ▸ MFN, seed rounds

The clause pays in software, not in deep tech



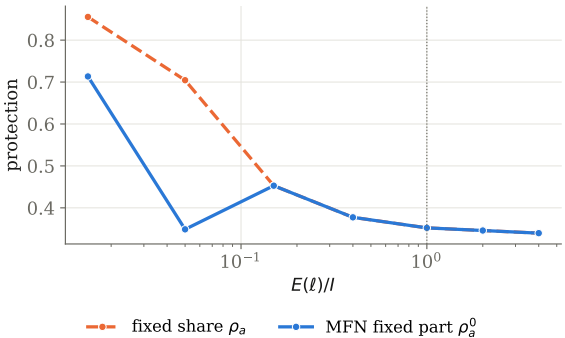
Liquidity shock: Gamma(2). Software: the MFN clause earns 1.5 to 5.4% more than the fixed share. Deep tech: the accelerator brings no money, so there is no MFN component, and the two protections earn the same.

The accelerator invests only when liquidity needs are small



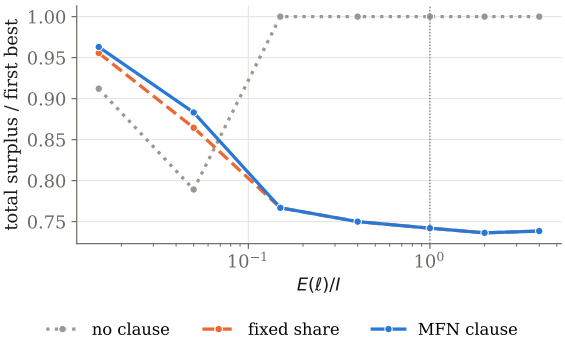
Liquidity shock: Gamma(2). Software, $E(\ell)/I \leq 0.05$: the accelerator invests, less with a protection, and almost all of its money goes through the clause. From $E(\ell)/I = 0.15$ on, its contribution vanishes under any protection.

Without a contribution, the protection is a fixed share



As σ_a tends to zero, so does the MFN amount, $m \leq \sigma_a$. The accelerator buys a fixed share with an amount and at a valuation that tend to zero.

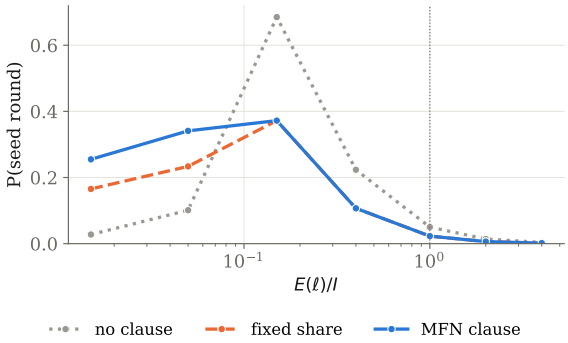
Welfare: total surplus



Software: MFN clause > fixed share > no clause. Beyond: no clause reaches the first best, and the protection loses about a quarter of it.

▸ Other shocks

Which projects survive: seed rounds



Software: the protection lowers the contribution and leaves more shocks to the seed investor. Beyond: it only shortens the seed investor's zone.

▸ Full funding ▸ Seed round bound

Takeaways

Main result

Any fixed share with a contribution is beaten by a small MFN component, which takes more from small rounds.

- The seed investor answers with an unpriced SAFE and invests beyond its need.
- In software, almost all of the money goes through the clause.
- Beyond software, the accelerator buys its share for almost no money, and this share costs a quarter of the surplus.

▸ Robustness ▸ Extensions

Thank you

Protection and Dilution:
SAFEs and MFN Clauses in Seed Financing

Stephen Dossou

Mines Paris PSL – Etilab

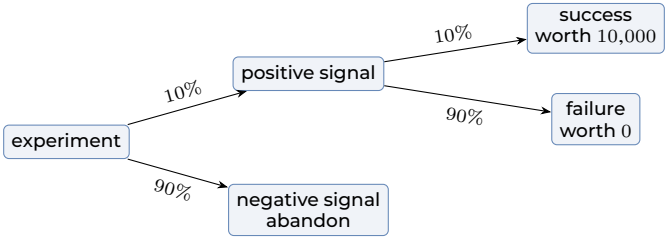
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An experiment makes a project worth funding



- Scaling costs 110. The project is worth $10{,}000$ with probability 1%: in one bet, its value is $100 - 110 = -10$, and no one funds it. This is (A1), $E(\pi) < I$.
- After a positive signal, it is worth $0.1 \times 10{,}000 - 110 = 890$.
- The experiment pays as long as it costs less than $10\% \times 890 = 89$.

Back

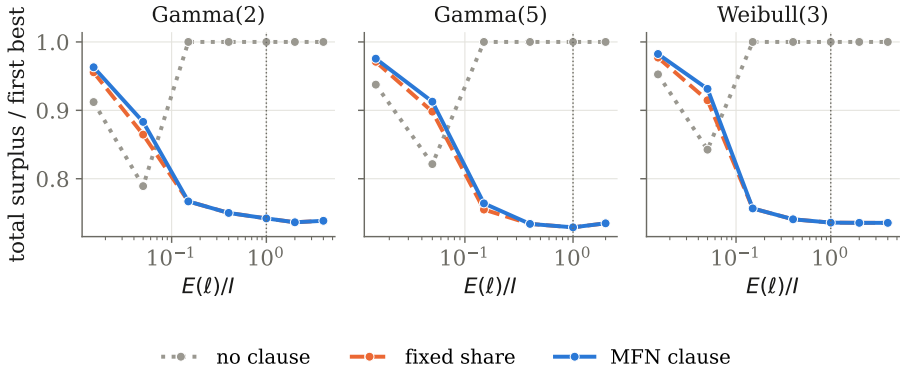
Robustness: another distribution of π

With a Beta$(2, 2)$ value on $[0, 1.8]$ instead of a uniform, nothing changes.

- Software: MFN clause $>$ fixed share $>$ no clause, almost all the money through the clause, total surplus 0.82–0.97 of the first best with the clause.
- Deep tech: the contribution vanishes, the fixed share is 0.18 to 0.42, and the total surplus falls to 0.74–0.76 of the first best.

Back

Robustness: other distributions of the liquidity shock



With a Gamma(5) or a Weibull(3) shock, the two regimes and the ranking are unchanged. Software: m^*/σ_a^* between 0.95 and 1. Deep tech: the total surplus falls to 0.73–0.76 of the first best.

Back Takeaways

Robustness: the founders keep a stake

The founders keep ρ_e before the seed round.

- The seed investor's answers hold with $\rho_e + \rho_a$ instead of ρ_a.
- Without protection, the first best is no longer reached.
- With $\rho_e = 0.5$, the optimal fixed share comes with a contribution that tends to zero, while the MFN clause keeps the accelerator investing: total surplus 0.79–0.87 of the first best, against 0.62–0.67.

Back

Extension: a priced SAFE against the MFN clause

If the accelerator could require a priced SAFE, a small MFN component would still beat any fixed share with a contribution.

The seed investor's share never exceeds $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$, whatever its SAFE, and the seed investor stops financing only rounds near the ends of its zone.

Proof Back

Extension: the accelerator's own SAFE as the fixed part

With its own SAFE as the fixed part, the optimal clause earns at most $2 \cdot 10^{-5}$ more than with a fixed share.

- The seed investor invests its need on almost every round, so only the value at the threshold I matters.
- Y Combinator used both forms of its 7%: a capped post-money SAFE from 2019, without MFN, and today a fixed 7% with the MFN SAFE.

Derivation Back

- The value $\pi \in [0,1]$ has density $f > 0$ and distribution F, with $E(\pi) < I$, the scaling cost. Before π is revealed, the project suffers a liquidity shock $\ell \geq 0$, of density $g > 0$ and distribution G. f and g are log-concave.
- The accelerator owns the whole project and brings σ_a before the shock, with a protection. This money is spent. If $\ell \leq \sigma_a$, it covers the shock alone.
- If $\ell > \sigma_a$, a seed investor observes the shock and makes a take-it-or-leave-it offer: it brings σ_s against a SAFE. If the accelerator rejects, the project is abandoned and it gets 0, so it accepts any round.
- The seed investor must cover the shock: (L) $\sigma_s \geq \ell - \sigma_a$.

Back

After the seed round, the project needs $x = I + \ell - \sigma_a - \sigma_s$. Once π is revealed, a competitive market brings x if $\pi > x$, at the pre-money value $\pi - x$. Otherwise the project is abandoned and returns nothing, including the money brought beyond the need. The equity held before the equity round is worth

$$W(x) = \int_x^1 (\pi - x)\, f(\pi)\, d\pi, \qquad W'(x) = -\big(1 - F(x)\big),$$

and $W(x) = E(\pi) - x$ for $x \le 0$. Since $\partial x / \partial \sigma_s = -1$, $W(x)$ has derivative $1 - F(x)$ in σ_s. Since $(\pi - I)^+ \le \pi$, $W(I) \le E(\pi) < I$.

Back

- A priced SAFE fixes the seed investor's share ρ_s at signing.
- A SAFE with cap V_s and discount $\lambda \in (0, 1]$ converts at the equity round at the lower of the cap and the discounted price: it claims $\sigma_s / \min(V_s, \lambda(\pi - x))$ of the equity held before the round. No cap: $V_s = +\infty$. No discount: $\lambda = 1$.
- A SAFE takes its claim as long as it fits in the equity the protection leaves. Beyond, the claims are scaled down in the same proportion. The protection comes first, as in YC's post-money SAFE.
- The fixed share ρ_a is a priced post-money SAFE: the accelerator buys ρ_a with σ_a, at the valuation σ_a / ρ_a. In the MFN clause, the fixed part ρ_a^0 is bought with $\sigma_a - m$. With no money, it is the limit of such SAFEs as the amount and the valuation tend to 0.

Back

On a shock $\ell \le \sigma_a$, the threshold is $I + \ell - \sigma_a$ and the accelerator keeps $W(I + \ell - \sigma_a)$. Let

$$B(\sigma_a) = \int_0^{\sigma_a} W(I + \ell - \sigma_a)\, g(\ell)\, d\ell - \sigma_a.$$

By Leibniz, with the boundary term $W(I)$ at $\ell = \sigma_a$,

$$\begin{aligned} B'(\sigma_a) &= W(I)\, g(\sigma_a) + \int_0^{\sigma_a} \big(1 - F(I + \ell - \sigma_a)\big) g(\ell)\, d\ell - 1 \\ &= W(I)\, g(\sigma_a) - \big(1 - G(\sigma_a)\big) - \int_0^{\sigma_a} F(I + \ell - \sigma_a)\, g(\ell)\, d\ell. \end{aligned}$$

B is continuous, zero at 0, and negative for $\sigma_a \ge I$: since $W(I + \ell - \sigma_a) \le E(\pi) - I + \sigma_a$, $B(\sigma_a) \le G(\sigma_a)(E(\pi) - I + \sigma_a) - \sigma_a < 0$.

Back

No clause, uncapped SAFE without discount

We show that the seed investor loses money at every amount

The SAFE claims $\sigma_s/(\pi - x)$ and takes at most the whole equity: in state $\pi > x$, it receives $\min(\sigma_s, \pi - x)$. It receives $\pi - x$ on $x < \pi < x + \sigma_s$, and σ_s above. With $F = 1$ above 1, its gross payoff is

$$\int_x^{x+\sigma_s} (\pi - x)\, f(\pi)\, d\pi + \sigma_s \big(1 - F(x + \sigma_s)\big).$$

Integrating the first term by parts, it equals $\sigma_s F(x + \sigma_s) - \int_x^{x+\sigma_s} F(\pi)\, d\pi$, and the gross payoff is $\sigma_s - \int_0^{\sigma_s} F(x + t)\, dt$. So

$$\Pi_s = -\int_0^{\sigma_s} F(x + t)\, dt < 0,$$

since $f > 0$. The SAFE is a debt of face value σ_s: it returns at most its amount. The seed investor finances no shock $\ell > \sigma_a$, and the accelerator chooses σ_a^B, as in the benchmark.

Back

The SAFE claims $\sigma_s/(\lambda(\pi - x))$ and receives $\min(\sigma_s/\lambda, \pi - x)$: a debt of face value σ_s/λ. Since $\min(\sigma_s/\lambda, \pi - x) = (\pi - x) - (\pi - x - \sigma_s/\lambda)^+$, and, for $y \geq 0$,

$$\int_x^1 (\pi - x - y)^+ f(\pi)\, d\pi = \int_{x+y}^1 \big(\pi - (x + y)\big) f(\pi)\, d\pi = W(x + y),$$

the gross payoff is $W(x) - W(x + \sigma_s/\lambda)$. Its derivative in the face value σ_s/λ is $1 - F(x + \sigma_s/\lambda) \geq 0$: the seed investor lowers λ.
Since $\pi \leq 1$, the equity never exceeds $1 - x$. Once the face value reaches $1 - x$, the seed investor receives the whole equity in every state, and earns $W(x) - \sigma_s$, decreasing in σ_s: $\sigma_s = \ell - \sigma_a$ and $x = I$.

Back

No clause, uncapped SAFE with discount
Synthesis

The discount $\lambda^* = \min\big(1, (\ell - \sigma_a)/(1 - I)\big)$ gives a face value of at least $1 - I$. The seed investor gets what the priced SAFE gives.

Result

Without clause, with an uncapped SAFE whose discount it chooses, the seed investor finances the shock if and only if $\ell - \sigma_a \le W(I)$. It brings $\sigma_s^* = \ell - \sigma_a$, chooses λ^*, takes the whole equity and earns $W(I) - (\ell - \sigma_a)$. The accelerator chooses σ_a^B.

Back

No clause, SAFE with a cap and a discount
Analysis and synthesis

The seed investor solves

$$\max_{\sigma_s, V_s, \lambda} \Pi_s = \int_x^1 \rho_s(\pi)(\pi - x) f(\pi)\, d\pi - \sigma_s, \qquad \rho_s(\pi) = \min\Big(\frac{\sigma_s}{\min(V_s, \lambda(\pi - x))}, 1\Big),$$

under (L).

Analysis. The gross payoff is at most $W(x)$, and $\Pi_s \le W(x) - \sigma_s$, which has derivative $-F(x) \le 0$ in σ_s and is at most $W(I) - (\ell - \sigma_a)$, at the need.

Synthesis. With $\sigma_s = \ell - \sigma_a$ and a cap $V_s \le \ell - \sigma_a$, the claim $(\ell - \sigma_a)/\min(V_s, \lambda(\pi - I))$ exceeds 1 in every state: the seed investor takes the whole equity and reaches the bound. The discount adds nothing. It gets what the priced SAFE gives, and the accelerator chooses σ_a^B.

Back

Fixed share, priced SAFE
The seed investor

The seed investor cannot take more than $1-\rho_a$. It solves

$$\max_{\sigma_s,\rho_s} \Pi_s = \rho_s W(x) - \sigma_s \quad \text{under (L)}, \quad \rho_s \le 1-\rho_a \text{ (P)}.$$

Analysis. Π_s rises in ρ_s, up to (P): $\rho_s = 1-\rho_a$. The profit $(1-\rho_a)W(x)-\sigma_s$ has derivative $(1-\rho_a)(1-F(x))-1<0$ in σ_s: (L) binds, $\sigma_s^* = \ell - \sigma_a$ and $x^* = I$.

Result

With a fixed share $\rho_a > 0$, the seed investor brings $\ell-\sigma_a$, takes $1-\rho_a$ and earns $(1-\rho_a)W(I)-(\ell-\sigma_a)$. It finances the shocks $\ell \le \sigma_a + (1-\rho_a)W(I)$, and the accelerator keeps $\rho_a W(I)$ on each.

Back

The accelerator solves

$$\max_{\sigma_a, \rho_a} \ \Pi_a = B(\sigma_a) + \rho_a W(I)\Big[G\big(\sigma_a + (1-\rho_a)W(I)\big) - G(\sigma_a)\Big].$$

Suppose (σ_a^*, ρ_a^*) optimal. At a given σ_a, the rent $\rho_a W(I)[G(\sigma_a + (1-\rho_a)W(I)) - G(\sigma_a)]$ is zero at $\rho_a = 0$ and at $\rho_a = 1$, and strictly positive in between. So ρ_a^* is interior, and the derivative in ρ_a, divided by $W(I)$, is zero:

$$\text{(i)} \quad G\big(\sigma_a + (1-\rho_a)W(I)\big) - G(\sigma_a) = \rho_a W(I)\, g\big(\sigma_a + (1-\rho_a)W(I)\big).$$

On the left, what a stronger protection brings on each seed project. On the right, what the accelerator kept on the last project, which leaves the zone.

Back

Fixed share, priced SAFE
The accelerator: the condition in σ_a

At a given ρ_a, the derivative of Π_a in σ_a is

$$B'(\sigma_a) + \rho_a W(I)\big[g(\sigma_a + (1-\rho_a)W(I)) - g(\sigma_a)\big]$$
$$= (1-\rho_a)W(I)\, g(\sigma_a) + \rho_a W(I)\, g\big(\sigma_a + (1-\rho_a)W(I)\big) - \big(1 - G(\sigma_a)\big) - \int_0^{\sigma_a} F(I + \ell - \sigma_a) g(\ell)\, d\ell.$$

One more dollar moves the whole seed zone by one dollar. The project $\ell = \sigma_a$ moves from the seed investor to the accelerator, which gains the share $1 - \rho_a$ there. A project enters at $\sigma_a + (1-\rho_a)W(I)$, where the accelerator keeps $\rho_a W(I)$. If $\sigma_a^* > 0$,

(ii) $\quad (1-\rho_a)W(I)g(\sigma_a) + \rho_a W(I) g\big(\sigma_a + (1-\rho_a)W(I)\big) = \big(1 - G(\sigma_a)\big) + \int_0^{\sigma_a} F(I + \ell - \sigma_a) g(\ell)\, d\ell.$

Back

(i) has a single solution in ρ_a. The rent is the product of ρ_a, log-concave, and of $G(\sigma_a + (1-\rho_a)W(I)) - G(\sigma_a)$. At $\ell > \sigma_a$, the log-derivative of $G(\ell) - G(\sigma_a)$ is $\left[\int_0^{\ell-\sigma_a} g(\ell - s)/g(\ell)\, ds\right]^{-1}$. As ℓ rises, the bound and each ratio rise, $\log g$ being concave: it decreases, and the factor is log-concave in ρ_a. The rent is strictly log-concave in ρ_a, and (i) gives its unique maximum $\rho_a^*(\sigma_a)$.
Against the benchmark. At a given σ_a, the derivative in σ_a exceeds $B'(\sigma_a)$ by $\rho_a W(I)[g(\sigma_a + (1-\rho_a)W(I)) - g(\sigma_a)]$, of either sign: the theory does not compare σ_a^* and σ_a^B. But at σ_a^B, any $\rho_a \in (0,1)$ adds a strictly positive rent, so $\Pi_a(\sigma_a^*, \rho_a^*) > B(\sigma_a^B)$.

Back

Fixed share, SAFE with a cap and a discount
Analysis and synthesis

The seed investor's share is its claim, capped by the protection:

$$\rho_s(\pi) = \min\Big(\frac{\sigma_s}{\min(V_s, \lambda(\pi - x))},\ 1 - \rho_a\Big).$$

Analysis. Its share never exceeds $1 - \rho_a$: $\Pi_s \le (1 - \rho_a)W(x) - \sigma_s$, the priced bound, maximal at the need, where $x = I$.
Synthesis. Take $\sigma_s = \ell - \sigma_a$ and a cap $V_s \le (\ell - \sigma_a)/(1 - \rho_a)$. In every state, $(\ell - \sigma_a)/\min(V_s, \lambda(\pi - I)) \ge (\ell - \sigma_a)/V_s \ge 1 - \rho_a$: the share is $1 - \rho_a$ everywhere, and the bound is reached. The discount is useless.
The capped SAFE gives the seed investor exactly what the priced SAFE gives, on the same projects. The accelerator's program, and (σ_a^*, ρ_a^*), are those of the fixed share with a priced SAFE.

Back

The accelerator's own capped SAFE
We show what the protection leaves

The accelerator protects its contribution with its own SAFE, of amount σ_a, cap $V_a \geq \sigma_a$ and discount λ_a. It claims $\sigma_a / \min(V_a, \lambda_a(\pi - x))$ and comes first. Since $1/\min(a,b) = \max(1/a, 1/b)$, the claim times $\pi - x$ is $\max(\sigma_a(\pi - x)/V_a, \sigma_a/\lambda_a)$. The protection takes this value, at most $\pi - x$. Since $\sigma_a/\lambda_a \leq V_a/\lambda_a$, three zones:

- $\pi - x \leq \sigma_a/\lambda_a$: it leaves 0;
- $\sigma_a/\lambda_a \leq \pi - x \leq V_a/\lambda_a$: it takes σ_a/λ_a and leaves $\pi - x - \sigma_a/\lambda_a$;
- $\pi - x \geq V_a/\lambda_a$: it takes the share σ_a/V_a and leaves $(1 - \sigma_a/V_a)(\pi - x)$.

In the three zones, this is $(\pi - x - \sigma_a/\lambda_a)^+ - (\sigma_a/V_a)(\pi - x - V_a/\lambda_a)^+$. With $\int_x^1 (\pi - x - y)^+ f = W(x+y)$, it leaves in expectation

$$W\big(x + \sigma_a/\lambda_a\big) - \frac{\sigma_a}{V_a} W\big(x + V_a/\lambda_a\big).$$

Back

Analysis. Its share does not exceed what the protection leaves: its profit is at most

$$W\big(x+\sigma_a/\lambda_a\big)-\frac{\sigma_a}{V_a}W\big(x+V_a/\lambda_a\big)-\sigma_s.$$

We show that this bound decreases in σ_s. Its derivative is

$$-F\big(x+\sigma_a/\lambda_a\big)-\frac{\sigma_a}{V_a}\Big(1-F\big(x+V_a/\lambda_a\big)\Big)\le 0.$$

It is maximal at the need, $x=I$.
Synthesis. A priced SAFE with $\rho_s=1$ receives everything the protection leaves. So does a SAFE with cap $V_s=\ell-\sigma_a$, whose claim is at least 1 in every state. Both reach the bound. The seed investor brings its need, and finances the shock if and only if

$$\ell-\sigma_a\le W\big(I+\sigma_a/\lambda_a\big)-\frac{\sigma_a}{V_a}W\big(I+V_a/\lambda_a\big).$$

Back

The accelerator's own capped SAFE
The values the SAFE can keep

On each seed project, the accelerator keeps $W(I) - W(I + \sigma_a/\lambda_a) + (\sigma_a/V_a)W(I + V_a/\lambda_a)$.
With the fixed share, the profit is the same function of the contribution and of the value kept, $\rho_a W(I)$ instead. The two contracts differ only by the values they allow, at a given contribution. With the fixed share: all of $[0, W(I)]$.
With the SAFE, if $\sigma_a = 0$ it keeps nothing. Suppose $\sigma_a > 0$.

- The protection leaves at most $W(I + \sigma_a/\lambda_a) \leq W(I + \sigma_a)$: the value kept is at least $W(I) - W(I + \sigma_a)$, approached as $\lambda_a = 1$ and $V_a \to +\infty$.
- It is at most $W(I)$, reached at $V_a = \sigma_a$.
- It is continuous in (V_a, λ_a) on a connected set: it takes every value in $\big(W(I) - W(I + \sigma_a), W(I)\big]$.

The minimum is $E\min(\sigma_a, (\pi - I)^+)$: the SAFE returns its amount when the project allows it.

Back

The accelerator's own capped SAFE
Synthesis: it never beats the fixed share

Let (σ_a^P, ρ_a^P) be an optimal fixed share, of profit Π_a^P.
The SAFE never does better. If $\sigma_a > 0$, the value it keeps is in $[0, W(I)]$, and the fixed share equal to this value divided by $W(I)$ keeps it with the same contribution: same profit, at most Π_a^P. If $\sigma_a = 0$, it keeps nothing.
It reaches Π_a^P if $\sigma_a^P > 0$ and $\rho_a^P W(I) > W(I) - W(I + \sigma_a^P)$. This value is then in the interval reached with the contribution σ_a^P.
It approaches Π_a^P if $\sigma_a^P = 0$. First, $\rho_a^P > 0$, since without money the fixed share earns $\rho_a W(I)[G((1-\rho_a)W(I)) - G(0)] > 0$ for $\rho_a \in (0,1)$. Next, $W(I) - W(I + \sigma_a) < \rho_a^P W(I)$ for σ_a small, so a SAFE of amount σ_a keeps exactly $\rho_a^P W(I)$. Last, as $\sigma_a \to 0$, $B(\sigma_a) \to 0$ and G is continuous: its profit tends to Π_a^P.

Back

The accelerator's own capped SAFE
A fixed share plus an uncapped SAFE

Suppose the accelerator combines a fixed share ρ_a and an uncapped SAFE without discount of amount u. We show that it earns exactly what a fixed share earns.

- In state $\pi > x$, the fixed share claims $\rho_a(\pi - x)$ and the SAFE the value u. The accelerator takes $\min(\pi - x, \rho_a(\pi - x) + u)$ and leaves $(1 - \rho_a)(\pi - x - u/(1 - \rho_a))^+$, that is $(1 - \rho_a)W(x + u/(1 - \rho_a))$ in expectation.
- The seed investor's bound $(1 - \rho_a)W(x + u/(1 - \rho_a)) - \sigma_s$ has derivative $(1 - \rho_a)(1 - F(\cdot)) - 1 < 0$: it brings its need.
- The accelerator keeps $W(I) - (1 - \rho_a)W(I + u/(1 - \rho_a))$ on every round, a value in $[0, W(I)]$.
- The fixed share equal to this value divided by $W(I)$ keeps it with the same contribution: same profit. Conversely, the fixed share alone is the case $u = 0$.

Back

MFN clause
The rule, and the share with a capped SAFE

The accelerator keeps at least ρ_a^0, and $m \le \sigma_a$ converts on the seed investor's terms.
The rule. Each SAFE claims a share of the equity held before the equity round. If the claims sum to at most $1 - \rho_a^0$, each gets its claim. Otherwise, each claim is multiplied by $1 - \rho_a^0$ over the sum. In both cases, each claim is multiplied by $\min\big(1, (1 - \rho_a^0)/\text{sum of the claims}\big)$.
Capped SAFE. Both SAFEs convert at the same price, $\min(V_s, \lambda(\pi - x))$. The seed investor claims $\sigma_s/\min(\cdot)$, the accelerator $m/\min(\cdot)$, and the sum is $(\sigma_s + m)/\min(\cdot)$. The seed investor's claim times $(1 - \rho_a^0)$ over the sum is $(1 - \rho_a^0)\sigma_s/(\sigma_s + m)$. By the rule, its share is

$$\min\Big(\frac{\sigma_s}{\min(V_s, \lambda(\pi - x))},\ \frac{(1 - \rho_a^0)\sigma_s}{\sigma_s + m}\Big).$$

Back

The seed investor converts at σ_s/ρ_s and claims ρ_s. The amount m converts at $\min(\sigma_s/\rho_s, \pi - x)$.

- States $\pi - x \geq \sigma_s/\rho_s$: the accelerator claims $m\rho_s/\sigma_s$, the sum is $\rho_s(\sigma_s + m)/\sigma_s$, and the share is $\min\big(\rho_s, (1-\rho_a^0)\sigma_s/(\sigma_s+m)\big)$.
- States $\pi - x < \sigma_s/\rho_s$: the accelerator claims $m/(\pi - x)$, the sum is $\rho_s + m/(\pi - x)$, and the share is

$$\min\Big(\rho_s,\ \frac{(1-\rho_a^0)\rho_s(\pi-x)}{\rho_s(\pi-x)+m}\Big).$$

We show that, if $m > 0$, this share is strictly below $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$. The function $t \mapsto t/(t+m)$ is strictly increasing, and $\rho_s(\pi - x) < \sigma_s$ in these states: $\rho_s(\pi-x)/(\rho_s(\pi-x)+m) < \sigma_s/(\sigma_s+m)$.
The ceiling. With both contracts, the share never exceeds $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$.

Back

MFN clause
We show what the accelerator keeps on a seed project

Take a shock the seed investor finances, with σ_s^*, $x^* = I + \ell - \sigma_a - \sigma_s^*$ and Π_s^*.

- The two investors share the value $\pi - x^*$ when $\pi > x^*$: $W(x^*)$ in expectation.
- The seed investor receives $\Pi_s^* + \sigma_s^*$. The accelerator keeps $W(x^*) - \sigma_s^* - \Pi_s^*$.
- Since $x^* \leq I$ and $W' = -(1 - F)$, $W(x^*) = W(I) + (I - x^*) - \int_{x^*}^{I} F(t)\,dt$, and $I - x^* = \sigma_s^* - (\ell - \sigma_a)$.

So the accelerator keeps

$$V = W(I) - (\ell - \sigma_a) - \Pi_s^* - \int_{x^*}^{I} F(t)\,dt :$$

the value at I, minus the round, minus the seed investor's profit, minus what the money beyond the need loses on the projects that are not fully funded. Π_s^* and x^* depend on ℓ only through the round $\ell - \sigma_a$.

Back

By the ceiling, the seed investor receives in expectation at most $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$ times $W(x)$:

$$\Pi_s \le \bar{\Pi}_s(\sigma_s) = \frac{(1-\rho_a^0)\sigma_s}{\sigma_s+m}W(x) - \sigma_s.$$

The ceiling has derivative $(1-\rho_a^0)m/(\sigma_s+m)^2$, and $W(x)$ has derivative $1-F(x)$ in σ_s. So

$$\bar{\Pi}_s'(\sigma_s) = (1-\rho_a^0)\left[\frac{mW(x)}{(\sigma_s+m)^2} + \frac{\sigma_s(1-F(x))}{\sigma_s+m}\right] - 1.$$

The first term is the gain in share: each extra dollar dilutes the MFN amount, and this gain is large when m weighs a lot against σ_s.
The bound stops rising at $x=0$. For $x \le 0$, $F(x)=0$ and $W(x) = E(\pi) - x < \sigma_s$, so $\bar{\Pi}_s' \le (1-\rho_a^0)\sigma_s(\sigma_s+2m)/(\sigma_s+m)^2 - 1 \le 0$. The bound reaches its maximum on $\ell - \sigma_a \le \sigma_s \le I + \ell - \sigma_a$.

Back

Large shocks. We show that the seed investor loses money at every amount when $\ell - \sigma_a + m > (1-\rho_a^0)W(I)$. Since $W(x) - W(I) = \int_x^I (1-F) \le I - x = \sigma_s - (\ell - \sigma_a)$,

$$(1-\rho_a^0)W(x) \le (1-\rho_a^0)W(I) + (1-\rho_a^0)\big(\sigma_s - (\ell - \sigma_a)\big) < \sigma_s + m.$$

Multiplying by $\sigma_s/(\sigma_s + m)$, the bound is negative.
At the need. At $\sigma_s = \ell - \sigma_a$, the bound is $(\ell - \sigma_a)\big[(1-\rho_a^0)W(I)/(\ell - \sigma_a + m) - 1\big]$, nonnegative if and only if $\ell - \sigma_a + m \le (1-\rho_a^0)W(I)$. The seed investor brings more than its need if the derivative at $x = I$ is positive:

$$(1-\rho_a^0)\Big[\frac{mW(I)}{(\ell - \sigma_a + m)^2} + \frac{(\ell - \sigma_a)(1-F(I))}{\ell - \sigma_a + m}\Big] > 1.$$

As $\ell - \sigma_a \to 0$, the left side tends to $(1-\rho_a^0)W(I)/m > 1$: the smallest rounds overinvest.

Back

MFN clause, capped SAFE
The seed investor: synthesis

Fix σ_s, take $\lambda = 1$ and the cap $V_s = (\sigma_s + m)/(1 - \rho_a^0)$. In every state, $\sigma_s / \min(V_s, \pi - x) \geq \sigma_s / V_s = (1 - \rho_a^0)\sigma_s/(\sigma_s + m)$: the share is the ceiling everywhere, and the profit is the bound. No (V_s, λ) gives more, so no discount helps. The bound, continuous on the bounded interval where it reaches its maximum, has a maximum.

Result

With an MFN clause and a capped SAFE, the seed investor finances the shock if and only if $\ell - \sigma_a + m \leq (1 - \rho_a^0)W(I)$. It takes $(1 - \rho_a^0)\sigma_s^*/(\sigma_s^* + m)$ with the cap $V_s^* = (\sigma_s^* + m)/(1 - \rho_a^0)$, and brings more than its need under the condition of the previous slide. The discount plays no role.

Back

The accelerator solves

$$\max_{\sigma_a,\rho_a^0,m\le\sigma_a} \Pi_a = B(\sigma_a) + \int_{\sigma_a}^{\bar{\ell}} \Big[W(I) - (\ell-\sigma_a) - \Pi_s^*(\ell) - \int_{x^*(\ell)}^{I} F(t)\,dt\Big] g(\ell)\,d\ell,$$

with $\bar{\ell} = \sigma_a + (1-\rho_a^0)W(I) - m$ the last shock the seed investor finances.
The last project. We show that the seed investor brings exactly its need there and earns 0. At $\bar{\ell}$, $\bar{\ell} - \sigma_a + m = (1-\rho_a^0)W(I)$, and the large-shock inequality, with equality instead of the strict hypothesis, gives $(1-\rho_a^0)W(x) \le \sigma_s + m$: the bound is at most 0 at every amount, and 0 at the need. Beyond the need, $x < I$, and since $F(I) > 0$, $W(x) - W(I) < I - x$: the bound is strictly negative. So $x^* = I$, it earns 0, and the accelerator keeps

$$W(I) - (\bar{\ell} - \sigma_a) = \rho_a^0 W(I) + m.$$

Back

The MFN component lowers the value kept on no financed project. The seed investor's share is at most $1-\rho_a^0$ and $x^* \leq I$: the accelerator keeps at least $\rho_a^0 W(I)$, what it keeps with $m=0$.
Take a fixed share (σ_a, ρ_a) with $\sigma_a > 0$ and $0 < \rho_a < 1$, not necessarily optimal. Keep σ_a and $\rho_a^0 = \rho_a$, and add a small $m > 0$. The seed zone becomes $0 < \ell - \sigma_a \leq (1-\rho_a^0)W(I) - m$. Fix a round $\delta \in \big(0, (1-\rho_a^0)W(I)\big)$ and take $m < (1-\rho_a^0)W(I) - \delta$: the round δ is in the zone before and after. It separates the small rounds, $\ell - \sigma_a < \delta$, where the seed investor may bring more than its need, from the others, where we show it brings its need. δ does not depend on m: we first let m tend to 0 at a given δ.

Back Back to the result

MFN clause, capped SAFE

On the rounds above δ, the seed investor brings its need

Large excesses lose. At $\sigma_s \geq \ell - \sigma_a$, the bound is at most $(1-\rho_a^0)W(x) - \sigma_s$, since $\sigma_s/(\sigma_s+m) \leq 1$. With $W(x) - W(I) = \int_x^I (1-F)$ and $I - x = \sigma_s - (\ell - \sigma_a)$,

$$(1-\rho_a^0)W(x) - \sigma_s = (1-\rho_a^0)W(I) - (\ell - \sigma_a) - \int_x^I \left[1 - (1-\rho_a^0)(1-F)\right],$$

whose integrand is at least ρ_a^0. At the need, the bound is $(1-\rho_a^0)W(I) - (\ell - \sigma_a) - (1-\rho_a^0)mW(I)/(\ell - \sigma_a + m)$. So the bound at σ_s is below the bound at the need as soon as $\sigma_s - (\ell - \sigma_a) > (1-\rho_a^0)mW(I)/(\rho_a^0\delta)$, which tends to 0 with m.
Small excesses lose too. Below this quantity, $x \geq 0$ and $W(x) \leq W(0)$ for small m: the derivative of the bound is at most $(1-\rho_a^0)[mW(0)/\delta^2 + 1 - F(x)] - 1$, which tends to $(1-\rho_a^0)(1-F(I)) - 1 < 0$, uniformly, F being continuous.

Back Back to the result

MFN clause, capped SAFE
The gain, the small rounds, the lost rounds

- On the rounds $\ell - \sigma_a \geq \delta$ of the zone, the seed investor brings its need, and the value kept gains

$$\left(1 - \frac{(1-\rho_a^0)(\ell-\sigma_a)}{\ell-\sigma_a+m}\right)W(I) - \rho_a^0 W(I) = \frac{(1-\rho_a^0)\,m\,W(I)}{\ell-\sigma_a+m}.$$

- On the rounds $\ell - \sigma_a < \delta$, the seed investor may bring more than its need, but the value kept does not fall: its share is at most $1-\rho_a^0$, and $W(x^*) \geq W(I)$.
- The lost projects are in a band of width m below the old last project, and the value kept on each was at most $W(I)$.

Back Back to the result

MFN clause, capped SAFE
The gain beats the loss, both divided by m

The loss. g, log-concave, is bounded: the loss is at most $mW(I)\sup g$. Divided by m, it stays below $W(I)\sup g$, whatever δ and m.
The gain. Divided by m, the gain on the rounds above δ is

$$(1-\rho_a^0)W(I)\int_{\sigma_a+\delta}^{\sigma_a+(1-\rho_a^0)W(I)-m}\frac{g(\ell)}{\ell-\sigma_a+m}\,d\ell.$$

As $m\to 0$, the upper bound tends to $\sigma_a+(1-\rho_a^0)W(I)$, and the integrand tends to $g(\ell)/(\ell-\sigma_a)$, staying below $\sup g/\delta$. The gain divided by m tends to

$$(1-\rho_a^0)W(I)\int_{\sigma_a+\delta}^{\sigma_a+(1-\rho_a^0)W(I)}\frac{g(\ell)}{\ell-\sigma_a}\,d\ell,$$

which grows without bound as δ falls, since $g(\sigma_a)>0$.
The choice of δ, then of m. Take δ such that this limit exceeds $W(I)\sup g$. At this δ, for m small enough, the gain divided by m still exceeds it: the profit strictly rises. Applied to the optimal fixed share, if it has a contribution: $m^*>0$.

◂ Back ◂ Back to the result

MFN clause, capped SAFE
The condition in σ_a

Fix (ρ_a^0, m) with $m < \sigma_a$. With the round $t = \ell - \sigma_a$, the seed integral is $\int_0^{(1-\rho_a^0)W(I)-m} V(t)\, g(\sigma_a + t)\, dt$, whose bound does not depend on σ_a. Its derivative is $\int V(t) g'(\sigma_a + t)\, dt$, and by parts

$$\int_0^{\bar{t}} (W(I) - t)\, g'(\sigma_a + t)\, dt = \big(\rho_a^0 W(I) + m\big) g(\bar{\ell}) - W(I) g(\sigma_a) + G(\bar{\ell}) - G(\sigma_a).$$

Adding $B'(\sigma_a)$, the derivative in σ_a is zero if and only if

$$\big(\rho_a^0 W(I) + m\big) g(\bar{\ell}) = \big(1 - G(\bar{\ell})\big) + \int_0^{\sigma_a} F(I + \ell - \sigma_a) g\, d\ell + \int_{\sigma_a}^{\bar{\ell}} \Big(\Pi_s^* + \int_{x^*}^{I} F\Big) g'(\ell)\, d\ell.$$

On the left, the last project, which enters the zone. On the right, the dollar lost on the shocks no one covers, the cost on the projects the accelerator covers alone, and the shift of the seed zone to shocks of another density, of no sign.

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MFN clause, capped SAFE

Why the dollar is lost only on the shocks no one covers

The accelerator pays its extra dollar on every shock: the $-\sigma_a$ of B. On a shock the seed investor finances, this dollar lowers the round $\ell - \sigma_a$ by one dollar, and the accelerator recovers only what a smaller round leaves it.

- With the fixed share, the seed investor's profit $(1-\rho_a)W(I) - (\ell - \sigma_a)$ rises by the whole dollar: the accelerator keeps $\rho_a W(I)$ whatever the round.
- With the MFN clause, on a round without overinvestment, it keeps $\rho_a^0 W(I) + (1-\rho_a^0)mW(I)/(\ell - \sigma_a + m)$, and recovers $(1-\rho_a^0)mW(I)/(\ell - \sigma_a + m)^2$ of the dollar: the component weighs more against a smaller round.

In the condition, $1 - G(\bar{\ell})$ counts the dollar lost only on the shocks no one covers, and the third term takes back what the seed investor's profit and the lost surplus gain on the smaller rounds.

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MFN clause, capped SAFE
The conditions in ρ_a^0 and in m

On the round t, let $V(t) = W(I) - t - \Pi_s^*(t) - \int_{x^*(t)}^{I} F$, and $\bar{t} = (1-\rho_a^0)W(I) - m$, where $V(\bar{t}) = \rho_a^0 W(I) + m$. Then $\Pi_a = B(\sigma_a) + \int_0^{\bar{t}} V(t) g(\sigma_a + t)\, dt$. ρ_a^0 and m act through the value kept on each round, and through the last round, which moves back by $W(I)$ per unit of ρ_a^0 and by one per unit of m:

$$\frac{\partial \Pi_a}{\partial \rho_a^0} = \int_0^{\bar{t}} \frac{\partial V}{\partial \rho_a^0} g(\sigma_a + t)\, dt - \big(\rho_a^0 W(I) + m\big) W(I)\, g(\bar{\ell}),$$

$$\frac{\partial \Pi_a}{\partial m} = \int_0^{\bar{t}} \frac{\partial V}{\partial m} g(\sigma_a + t)\, dt - \big(\rho_a^0 W(I) + m\big) g(\bar{\ell}).$$

Without overinvestment, $V(t) = \rho_a^0 W(I) + (1-\rho_a^0) m W(I)/(t+m)$, so $\partial V/\partial \rho_a^0 = tW(I)/(t+m)$ and $\partial V/\partial m = (1-\rho_a^0) t W(I)/(t+m)^2$. With interior overinvestment, $\partial V/\partial \theta = -\partial \Pi_s/\partial \theta - F(x^*)\, \partial \sigma_s^*/\partial \theta$.

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MFN clause, capped SAFE
The conditions at the optimum

The constraint is $0 \le m \le \sigma_a$. Let D_σ be the derivative in σ_a at given (ρ_a^0, m).

- If $0 < \rho_a^0 < 1$: $\partial\Pi_a/\partial\rho_a^0 = 0$.
- If $0 < m < \sigma_a$: $\partial\Pi_a/\partial m = 0$ and $D_\sigma = 0$.
- If $m = \sigma_a > 0$: $D_\sigma \le 0$, $\partial\Pi_a/\partial m \ge 0$, and $D_\sigma + \partial\Pi_a/\partial m = 0$.

The combination without the last project.

$$W(I)\frac{\partial\Pi_a}{\partial m} - \frac{\partial\Pi_a}{\partial\rho_a^0} = \int_0^{\bar{t}} \left[W(I)\frac{\partial V}{\partial m} - \frac{\partial V}{\partial\rho_a^0}\right] g(\sigma_a + t)\, dt.$$

Without overinvestment, the integrand is $tW(I)(\bar{t} - t)/(t + m)^2 > 0$. With overinvestment, it contains the cost $F(x^*)$ per extra dollar. If $m < \sigma_a$, the combination is zero. If $m = \sigma_a$, it is nonnegative: the gain wins, and $m \le \sigma_a$ stops the transfer.

Back

Analysis. The share never exceeds $(1-\rho_a^0)\sigma_s/(\sigma_s+m)$: the profit has the same bound $\bar{\Pi}_s$ as with the capped SAFE, which reaches it. It earns at most as much, and finances no shock with $\ell - \sigma_a + m > (1-\rho_a^0)W(I)$.
Strictly less if $m > 0$. The priced profit is continuous in (σ_s, ρ_s) on a compact set and reaches a maximum, at most the maximum of $\bar{\Pi}_s$. Suppose they are equal, at σ_s^p. Then the profit is $\bar{\Pi}_s(\sigma_s^p)$. But the states $x < \pi < x + \sigma_s^p/\rho_s$ have positive probability, and there the share is strictly below the ceiling when $m > 0$: the profit is strictly below $\bar{\Pi}_s(\sigma_s^p)$, a contradiction. If $m = 0$, the two contracts give the same profit.

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MFN clause, priced SAFE

The priced seed investor still finances almost all the zone

Start from a fixed share (σ_a, ρ_a^0), $\sigma_a > 0$, $0 < \rho_a^0 < 1$, add a small m, and fix δ as before. On the round $\ell - \sigma_a$, consider the contract that brings the need, $x = I$, against

$$\rho_s = \frac{(1-\rho_a^0)(\ell-\sigma_a)}{\ell-\sigma_a+m}, \quad \text{so that} \quad 1-\rho_a^0-\rho_s = \frac{(1-\rho_a^0)m}{\ell-\sigma_a+m}.$$

In the states $\pi - I \geq (\ell - \sigma_a + m)/(1-\rho_a^0)$, its share is ρ_s. In the others, it takes $(1-\rho_a^0)\rho_s(\pi-I)/(\rho_s(\pi-I)+m)$, and loses relative to ρ_s

$$\frac{\rho_s m}{\rho_s(\pi-I)+m}\left[1-\frac{(1-\rho_a^0)(\pi-I)}{\ell-\sigma_a+m}\right].$$

Times $\pi - I$, this is at most m. So it receives at least $\rho_s(\pi-I)-m$, and its profit is at least

$$\frac{(1-\rho_a^0)W(I)(\ell-\sigma_a)}{\ell-\sigma_a+m} - (\ell-\sigma_a) - m.$$

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MFN clause, priced SAFE
The rounds where this lower bound is positive

The lower bound is nonnegative if and only if $(1-\rho_a^0)W(I)(\ell-\sigma_a) \geq (\ell-\sigma_a+m)^2$, that is

$$\sqrt{(1-\rho_a^0)W(I)(\ell-\sigma_a)} - (\ell-\sigma_a) - m \geq 0.$$

The left side is concave in $\ell-\sigma_a$. Suppose $m \leq (1-\rho_a^0)W(I)/4$.

- At $\ell-\sigma_a = 4m^2/((1-\rho_a^0)W(I))$, the square root is $2m$, and the left side is $m - 4m^2/((1-\rho_a^0)W(I)) \geq 0$.
- At $\ell-\sigma_a = (1-\rho_a^0)W(I) - 3m$, it is nonnegative if and only if $(1-\rho_a^0)W(I)m \geq 4m^2$, true.

A concave function nonnegative at two points is nonnegative between them. The priced seed investor, whose optimal profit is at least that of this contract, finances all the rounds between these two values.

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MFN clause, priced SAFE

Above δ, the value kept gains at least as with the capped SAFE

By the ceiling, Π_s^p is at most $(1-\rho_a^0)\sigma_s^p/(\sigma_s^p+m)$ times $W(x^p)$, minus σ_s^p, with $W(x^p) = W(I) + \sigma_s^p - (\ell - \sigma_a) - \int_{x^p}^{I} F$. Since F rises, $\int_{x^p}^{I} F \le F(I)(\sigma_s^p - (\ell - \sigma_a))$, and

$$\Pi_s^p + \int_{x^p}^{I} F \le \frac{(1-\rho_a^0)\sigma_s^p}{\sigma_s^p + m}\left[W(I) + \sigma_s^p - (\ell - \sigma_a)\right] - \sigma_s^p + \left(1 - \frac{(1-\rho_a^0)\sigma_s^p}{\sigma_s^p + m}\right)F(I)\left(\sigma_s^p - (\ell - \sigma_a)\right).$$

Also $x^p \ge 0$, so $\sigma_s^p - (\ell - \sigma_a) \le I$. The derivative of the bound in σ_s^p is at most $(1-\rho_a^0)m(W(I)+I)/\delta^2 - \rho_a^0(1-F(I))$, negative for small m. The bound is maximal at the need, and the value kept is at least

$$\rho_a^0 W(I) + \frac{(1-\rho_a^0)\, m\, W(I)}{\ell - \sigma_a + m}.$$

Back

MFN clause, priced SAFE
The gain beats the loss

The small rounds. On the financed rounds below δ, the accelerator's share does not fall below ρ_a^0: the value kept does not fall.
The loss. The lost rounds lie in two bands, of widths $4m^2/((1-\rho_a^0)W(I))$ and $3m$, at most $4m$ in all. The accelerator kept at most $W(I)$ on each, and g is bounded: the loss divided by m stays below $4W(I)\sup g$.
The gain. Divided by m, it is at least

$$(1-\rho_a^0)W(I)\int_{\sigma_a+\delta}^{\sigma_a+(1-\rho_a^0)W(I)-3m}\frac{g(\ell)}{\ell-\sigma_a+m}\,d\ell,$$

which tends, as $m\to 0$, to $(1-\rho_a^0)W(I)\int_{\sigma_a+\delta}^{\sigma_a+(1-\rho_a^0)W(I)} g(\ell)/(\ell-\sigma_a)\,d\ell$, unbounded as δ falls.
The choice of δ, then of m. Take δ such that this limit exceeds $4W(I)\sup g$, then m small enough: the profit strictly rises. Applied to the optimal fixed share, if it has a contribution: $m^*>0$.

Back

MFN clause with the accelerator's own SAFE
The seed investor

The accelerator brings σ_a: m converts on the seed investor's terms, and $\sigma_a - m$ on its own SAFE, of cap $V_a \geq \sigma_a - m$ and discount λ_a. Its protection comes first, then the seed investor and the MFN amount share what it leaves, which is in expectation

$$W\Big(x + \frac{\sigma_a - m}{\lambda_a}\Big) - \frac{\sigma_a - m}{V_a} W\Big(x + \frac{V_a}{\lambda_a}\Big).$$

The seed investor's share never exceeds $\sigma_s/(\sigma_s + m)$ times what the protection leaves, and a cap $V_s = (\sigma_s + m)/(1 - (\sigma_a - m)/V_a)$ reaches this ceiling in every state. Its profit is at most $\sigma_s/(\sigma_s + m)$ times the bracket, minus σ_s. The bracket has derivative between 0 and 1 in σ_s, so the large-shock argument applies: the seed investor finances the shock if and only if

$$\ell - \sigma_a + m \leq W\Big(I + \frac{\sigma_a - m}{\lambda_a}\Big) - \frac{\sigma_a - m}{V_a} W\Big(I + \frac{V_a}{\lambda_a}\Big).$$

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MFN clause with the accelerator's own SAFE
The comparison with the fixed share

At the threshold. At given (σ_a, m), the fixed part ρ_a^0 leaves $(1-\rho_a^0)W(I)$ at I. If it leaves as much as the SAFE, the two contracts have the same zone, and, where the seed investor brings its need, the accelerator keeps the same value. They differ only on the small rounds, where the seed investor overinvests and what the protection leaves below I matters.

The fixed share at the limit. Fix $\rho_a^0 \in (0,1)$, take $\lambda_a = 1$ and $V_a = (\sigma_a - m)/\rho_a^0$, and let $m \to \sigma_a$. What the protection leaves is $W(x+\sigma_a-m) - \rho_a^0 W(x + (\sigma_a - m)/\rho_a^0)$, and W is 1-Lipschitz: it is within $2(\sigma_a - m)$ of $(1-\rho_a^0)W(x)$, uniformly in x. The seed investor's bound, and its maximal profit, converge to those of the MFN clause with fixed part ρ_a^0 and $m = \sigma_a$.

What the priority value changes. It acts only on the small rounds. The theory does not say whether it strictly raises the accelerator's profit: the simulations measure it.

Back

A fixed share is an MFN clause with $\rho_a^0 = \rho_a$ and $m = 0$, and no clause one with $\rho_a^0 = m = 0$. In all cases, the accelerator covers $[0, \sigma_a]$ alone, and the seed investor finances $\sigma_a < \ell \leq \bar{\ell} = \sigma_a + (1 - \rho_a^0)W(I) - m$. The project is fully funded with probability

$$\int_0^{\sigma_a} \big(1 - F(I + \ell - \sigma_a)\big) g(\ell)\, d\ell + \int_{\sigma_a}^{\bar{\ell}} \big(1 - F(x^*)\big) g(\ell)\, d\ell.$$

When the seed investor covers only its need, the second term is $(1 - F(I))[G(\bar{\ell}) - G(\sigma_a)]$. The derivative in $\bar{\ell}$ is $(1 - F(I))g(\bar{\ell}) > 0$, and in σ_a, at a given $\bar{\ell}$, $\int_0^{\sigma_a} f(I + \ell - \sigma_a) g(\ell)\, d\ell$. With a fixed share, $\bar{\ell}$ rises one for one with σ_a, so

$$\frac{dP}{d\sigma_a} = \int_0^{\sigma_a} f(I + \ell - \sigma_a) g(\ell)\, d\ell + \big(1 - F(I)\big) g\big(\sigma_a + (1 - \rho_a)W(I)\big) > 0.$$

Back

The probability changes by

$$\underbrace{\int_{\sigma_a^B}^{\sigma_a^P}\left[\int_0^s f(I+\ell-s)g(\ell)\,d\ell+\big(1-F(I)\big)g\big(s+W(I)\big)\right]ds}_{\text{contribution}}$$

$$\underbrace{-\big(1-F(I)\big)\Big[G\big(\sigma_a^P+W(I)\big)-G\big(\sigma_a^P+(1-\rho_a^P)W(I)\big)\Big]}_{\text{last project}\,>0}.$$

Proof. Go from $(\sigma_a^B, 0)$ to (σ_a^P, ρ_a^P) in two steps. Raising the contribution from σ_a^B to σ_a^P without protection integrates $dP/d\sigma_a$ at $\rho_a = 0$: the first term, of the sign of $\sigma_a^P - \sigma_a^B$. Raising the share from 0 to ρ_a^P at σ_a^P moves the last shock down from $\sigma_a^P + W(I)$ to $\sigma_a^P + (1-\rho_a^P)W(I)$: the second term, strictly negative. If the fixed share does not raise the contribution, it strictly lowers full funding.

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With a capped SAFE, the probability changes by

$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^M}\Big[\int_0^s f(I+\ell-s)g\,d\ell+\big(1-F(I)\big)g\big(s+(1-\rho_a^P)W(I)\big)\Big]ds}_{\text{contribution}}$$

$$+\underbrace{\big(1-F(I)\big)\Big[G(\bar\ell^M)-G\big(\sigma_a^M+(1-\rho_a^P)W(I)\big)\Big]}_{\text{last project}}+\underbrace{\int_{\sigma_a^M}^{\bar\ell^M}\big(F(I)-F(x^*)\big)g(\ell)\,d\ell}_{\text{beyond the need}\ \geq\ 0}.$$

Proof. Raise the contribution from σ_a^P to σ_a^M at the share ρ_a^P, which integrates $dP/d\sigma_a$, then move the last shock from $\sigma_a^M+(1-\rho_a^P)W(I)$ to $\bar\ell^M$. The third term is nonnegative since $x^*\leq I$. The first has the sign of $\sigma_a^M-\sigma_a^P$, the second that of $\rho_a^P W(I)-\rho_a^{0M}W(I)-m^M$. The theory signs only the third.

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Seed rounds
A seed round covers at most $W(I) < I$

Whatever the protection and the SAFE, the seed investor finances a shock $\ell > \sigma_a$ only if $\ell - \sigma_a \leq W(I) < I$.

Proof. What the seed investor receives is at most the equity, $W(x)$: $\Pi_s \leq W(x) - \sigma_s$, which decreases in σ_s and is at most $W(I) - (\ell - \sigma_a)$ at the need. And $W(I) < I$ since $E(\pi) < I$.

The probabilities. A seed round happens with probability $G(\sigma_a + (1 - \rho_a^0)W(I) - m) - G(\sigma_a)$:

- no clause, with any SAFE that finances: $G(\sigma_a^B + W(I)) - G(\sigma_a^B)$;
- fixed share: $G(\sigma_a^P + (1 - \rho_a^P)W(I)) - G(\sigma_a^P)$;
- MFN clause, capped SAFE: $G(\bar{\ell}^M) - G(\sigma_a^M)$;
- MFN clause, priced SAFE: the priced seed investor earns less, and finances only shocks the capped one finances.

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From no clause to the optimal fixed share, the probability of a seed round changes by

$$\underbrace{\left[G(\sigma_a^P + W(I)) - G(\sigma_a^P)\right] - \left[G(\sigma_a^B + W(I)) - G(\sigma_a^B)\right]}_{\text{contribution}}$$

$$- \underbrace{\left[G(\sigma_a^P + W(I)) - G(\sigma_a^P + (1-\rho_a^P)W(I))\right]}_{\text{last project} > 0}.$$

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Seed rounds
From the fixed share to the MFN clause

From the fixed share to the MFN clause, the probability of a seed round changes by

$$\underbrace{\left[G(\sigma_a^M + (1-\rho_a^P)W(I)) - G(\sigma_a^M)\right] - \left[G(\sigma_a^P + (1-\rho_a^P)W(I)) - G(\sigma_a^P)\right]}_{\text{contribution}}$$

$$+ \underbrace{G(\bar{\ell}^M) - G\big(\sigma_a^M + (1-\rho_a^P)W(I)\big)}_{\text{last project}}.$$

Another contribution slides the seed zone, which keeps its width: the probability changes as g is higher at one end or the other. What the accelerator keeps on the last project shortens the zone. The theory signs only the last term of the first move.

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Total surplus
The definition and the surplus of a covered project

Total surplus is the sum of the agents' expected profits: what each receives in expectation minus the money it brings. The equity investor earns zero, and the entrepreneur sold the project. Transfers cancel, so on a covered project of threshold x it is the value of the equity minus all the money brought before the equity round, $W(x) - (I + \ell - x)$. On a project abandoned for lack of a seed round, it is $-\sigma_a$.
We show that a covered project has surplus $W(I) - \ell - \int_x^I F(t)\,dt$. Since $W' = -(1 - F)$,

$$W(x) = W(I) + (I - x) - \int_x^I F(t)\,dt,$$

and we subtract $I + \ell - x$. By parts,

$$\int_x^I F(t)\,dt = (I - x)F(x) + \int_x^I (I - \pi)f(\pi)\,d\pi :$$

the money beyond the need is lost when $\pi < x$, and funds projects worth less than I when $x < \pi < I$.

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Total surplus
The first best and the four losses

A covered project has surplus at most $W(I) - \ell$, at $x = I$, and an abandoned one at most 0. So

$$TS \le \int_0^{W(I)} \big(W(I) - \ell\big) g(\ell)\, d\ell = TS^*,$$

reached with $\sigma_a = 0$, covering exactly the shocks $\ell \le W(I)$ at I. Summing the surpluses,

$$TS = TS^* - L_{acc} - L_{cov} - L_{over} - L_{sunk},$$

$$L_{acc} = \int_0^{\sigma_a} \int_{I+\ell-\sigma_a}^{I} F\, dt\, g\, d\ell, \quad L_{cov} = \int_{\bar{\ell}}^{W(I)} \big(W(I) - \ell\big) g\, d\ell, \quad L_{over} = \int_{\sigma_a}^{\bar{\ell}} \int_{x^*}^{I} F\, dt\, g\, d\ell, \quad L_{sunk} = \sigma_a\big(1 - G$$

The sum is $\int_0^{\bar{\ell}} (W(I) - \ell) g - L_{acc} - L_{over} - \sigma_a(1 - G(\bar{\ell}))$, and $\int_0^{\bar{\ell}} (W(I) - \ell) g = TS^* - L_{cov}$. $L_{cov} \ge 0$ either way: below $W(I)$, positive shocks left uncovered; above, negative shocks covered because the contribution is spent.

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Total surplus
The contribution and the last project

When the seed investor covers only its need, the surplus is $\int_0^{\bar{\ell}} (W(I) - \ell)g - L_{acc}(\sigma_a) - \sigma_a(1 - G(\bar{\ell}))$. Its derivative in $\bar{\ell}$ is $(\sigma_a + W(I) - \bar{\ell})g(\bar{\ell}) = (\rho_a^0 W(I) + m)g(\bar{\ell})$. One more dollar, at a given value kept on the last project, raises $\bar{\ell}$ by one and changes the surplus by

$$\big(\rho_a^0 W(I) + m\big)g(\bar{\ell}) - \big(1 - G(\bar{\ell})\big) - \int_0^{\sigma_a} F(I + \ell - \sigma_a)g(\ell)\, d\ell.$$

Without clause, this is negative: the surplus falls with the contribution.

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Total surplus
From no clause to the optimal fixed share

From no clause to the optimal fixed share, the surplus changes by

$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^B} \Big[\big(1 - G(s + W(I))\big) + \int_0^s F(I + \ell - s)g \, d\ell\Big] ds}_{\text{contribution, sign of } \sigma_a^B - \sigma_a^P}$$

$$- \underbrace{\int_{\sigma_a^P + (1-\rho_a^P)W(I)}^{\sigma_a^P + W(I)} \big(\sigma_a^P + W(I) - \ell\big)g \, d\ell}_{\text{last project} \, > \, 0}.$$

Proof. Lower the contribution from σ_a^B to σ_a^P without protection, which integrates the derivative of the previous slide, then raise the share from 0 to ρ_a^P: the last shock moves down from $\sigma_a^P + W(I)$ to $\sigma_a^P + (1 - \rho_a^P)W(I)$, and each shock in between loses its value $\sigma_a^P + W(I) - \ell$.

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With a capped SAFE, the surplus changes by

$$\underbrace{\int_{\sigma_a^P}^{\sigma_a^M} \left[\rho_a^P W(I) g\big(s + (1-\rho_a^P)W(I)\big) - \big(1 - G(s + (1-\rho_a^P)W(I))\big) - \int_0^s F(I+\ell-s)g\, d\ell\right] ds}_{\text{contribution}}$$

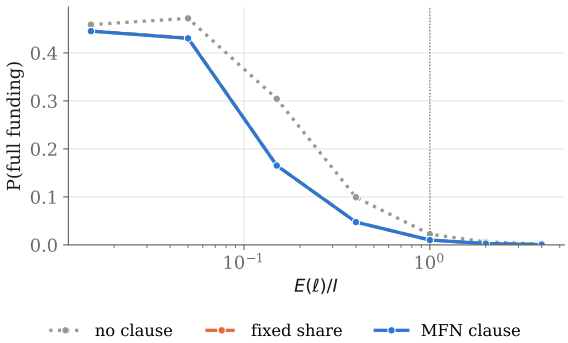
$$+ \underbrace{\int_{\sigma_a^M + (1-\rho_a^P)W(I)}^{\bar{\ell}^M} \big(\sigma_a^M + W(I) - \ell\big) g(\ell)\, d\ell}_{\text{last project}} - \underbrace{\int_{\sigma_a^M}^{\bar{\ell}^M} \int_{x^*}^{I} F(t)\, dt\; g(\ell)\, d\ell}_{\text{beyond the need } \geq 0}.$$

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Proof. Raise the contribution from σ_a^P to σ_a^M at the share ρ_a^P, which integrates the derivative of the contribution, then move the last shock to $\bar{\ell}^M$, each added shock being worth $\sigma_a^M + W(I) - \ell$. L_{over} exists only with the MFN clause. The second term has the sign of $\rho_a^P W(I) - \rho_a^{0M} W(I) - m^M$, the third is negative, the first has no sign. The money beyond the need raises full funding but lowers the surplus: the projects it pushes over the threshold are worth less than I.

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Which projects survive: full funding



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